AI BASED MICROCLIMATE PREDICTION SYSTEM FOR ROOFTOP FARMING

ABSTRACT:-

As an agriculture is the backbone of India and contributes around 18% of India's overall GDP, also in the era of technology, most of the sectors like health-care, civil, defense and agriculture are rapidly expanding with features of Artificial Intelligence (AI) and Machine Learning (ML). Also, rooftop farming is growing as a carbon friendly farming practice making use of unused space of building rooftops. Agricul turalists can struggle due to the traditional approaches of crop management that can cause a decrease in crop yield. Microclimate prediction system with the help of arti f icial intelligence and features of IoT sensors for rooftop farms is an interesting topic for research in field of technology. The main intention of this work is to build the system that will help to protect the crops from the unfavourable microclimatic factor that differ from the regional climatic conditions. Microclimatic factors like tempera ture, humidity, moisture can be monitored with helps of Internet of Things (IoT) sen sors which monitor the crop health with the features of Artificial Intelligence which make it more efficient and predictable. The proposed system will contain network of wireless sensors which will are able for real-time monitoring of the data which can be used by Machine learning models like Random Forest and Long Short-Term Memory (LSTM) for prediction. This predicted data will be evaluated by different performance metrics like Root Mean Square Error (RMSE), Accuracy, etc. Finally this work aims to highlight an approach that how Artificial Intelligence can be used in field of modern farming which helps in sustainable growth with accurate predic tion of microclimate. The proposed system will work on ground realities and not only help the farmers but also encourage people to lead toward organic farming, will promote the culture of healthy lifestyle using modern tools of artificial intelligence. Keywords: Artificial Intelligence, Microclimate Prediction, Rooftop Farming, IoT Sensors, Machine Learning, Urban Agriculture